

Ex 10: Data Visualization for an Inventory Management System

Aim

To develop a web-based dashboard that visualizes inventory data using graphical representations (Pie Chart and Bar Chart) to facilitate better stock management and analysis.

Procedure

The project was implemented using a combination of web technologies and an external data visualization library:

1. HTML Structure:

- Defined two `<canvas>` elements with unique IDs (pieChart and barChart) to serve as containers for the graphics.
- Imported the **Chart.js** library via a CDN link to enable advanced charting capabilities.

2. Data Modeling (JavaScript):

- Created a central data object (inventoryData) containing labels for various categories: Electronics, Clothing, Home Appliances, Books, and Toys.
- Assigned specific numerical values to these categories to represent "Items in Stock."

3. Graphic Rendering:

- Used the `getContext('2d')` method to target the HTML canvases.
- Instantiated two new Chart objects:
 - **Pie Chart:** Configured to show the proportional distribution of inventory.
 - **Bar Chart:** Configured to show a direct numerical comparison across categories with the Y-axis starting at zero.

4. Styling: * Applied CSS to center-align the content and provide adequate spacing for a clean, professional dashboard appearance.

```

index.html > html > head > style
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Inventory Management Visualization</title>
7  <style>
8  body {
9    font-family: Arial, sans-serif;
10   text-align: center;
11   margin: 50px;
12 }
13 canvas {
14   margin: 20px auto;
15 }
16
17 </style>
18 </head>
19 <body>
20 <h1>Inventory Management System</h1>
21 <canvas id="pieChart" width="400" height="400"></canvas>
22 <canvas id="barChart" width="400" height="400"></canvas>
23 <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
24 <script src="script.js"></script>
25 </body>
26 </html>
27
28

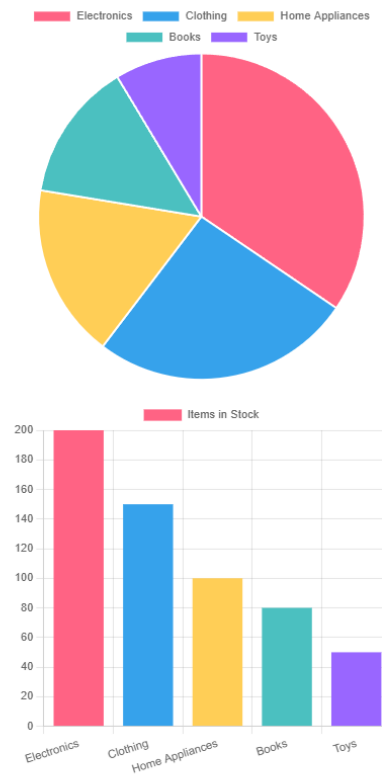
```

```

JS script.js
1  const inventoryData = {
2    labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
3    datasets: [
4      {
5        label: 'Items in Stock',
6        data: [200, 150, 100, 80, 50],
7        backgroundColor: [
8          '#FF6384',
9          '#36A2EB',
10         '#FFC566',
11         '#4DC0C0',
12         //
13         '#9966FF'
14       ],
15     },
16   ],
17 },
18 };
19
20 // Creating the Pie Chart
21 const ctxPie = document.getElementById('pieChart').getContext('2d');
22 const pieChart = new Chart(ctxPie, {
23   type: 'pie',
24   data: inventoryData,
25   options: {
26     responsive: true,
27     title: {
28       display: true,
29       text: 'Inventory Distribution'
30     },
31   },
32 });
33
34 // Creating the Bar Chart
35 const ctxBar = document.getElementById('barChart').getContext('2d');
36 const barChart = new Chart(ctxBar, {
37   type: 'bar',
38   data: inventoryData,
39   options: {
40     responsive: true,
41     title: {
42       display: true,
43       text: 'Items in Stock by Category'
44     },
45     scales: {
46       yAxes: [{
47         ticks: {
48           beginAtZero: true
49         }
50       }
51     ]
52   },
53 },
54 });

```

Inventory Management System



Results

The application successfully transformed raw numerical data into interactive visual components:

- **Proportional Analysis:** The **Pie Chart** clearly displays that "Electronics" occupies the largest segment of the inventory.
- **Comparative Analysis:** The **Bar Chart** provides an easy-to-read comparison, showing a downward trend in stock levels from Electronics (200 units) to Toys (50 units).
- **User Interface:** The final output (as seen in the screenshot) presents a responsive and color-coded "Inventory Management System" dashboard that is easy for a user to interpret at a glance.

